

Shiyu (Rayne) Xu



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Research Goal

Human-Computer Interaction (HCI), Human-AI Collaboration. I aim to design and develop intelligent systems that enhance human well-being and support better sense-making in both daily and clinical contexts.

Education

2020 – 2022	University of Michigan – Ann Arbor M.S. in Information (HCI)	Ann Arbor, MI, USA GPA: 4.0/4.0
2016 – 2020	University of Wisconsin – Madison B.S. in Computer Science Minors: Graphic Design & Mathematics	Madison, WI, USA GPA: 3.7/4.0

Publications

*= equal contribution

Peer Reviewed Publications & Manuscripts

- [P3] MIND: Empowering Mental Health Clinicians with Multimodal Data Insights through a Narrative Dashboard 
Shiyu Xu*, Ruishi Zou*, Margaret E. Morris, Jihan Ryu, Timothy D. Becker, Nicholas Allen, Anne Marie Albano, Randy Auerbach, Dan Adler, Varun Mishra, Lace Padilla, Dakuo Wang, Ryan Sultan, Xuhai “Orson” Xu
CHI 2026 – *Conditionally Accepted*
- [P2] How Pairing by Code Similarity Influences Discussions in Peer Learning 
Shiyu Xu, Ashley Ge Zhang, Steve Oney
CHI 2023 (LBW) – *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*
- [P1] “It would probably turn into a social faux-pas”: Users’ and Bystanders’ Preferences of Privacy Awareness Mechanisms in Smart Homes. 
Parth Kirankumar Thakkar, Shijing He, **Shiyu Xu**, Danny Yuxing Huang, Yaxing Yao
CHI 2022 – *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*

Research Experience

2025 – Present	Research on Mental Health Interfaces Bridging the Patient–Provider Journey Advisor: Dr. Dan Adler Research Focus: How AI-driven mental health systems can enhance healthcare providers’ efficiency and decision making while enabling transparent sharing of data across patients and providers. - Designed patient-facing data-sharing experiences that integrate multi-modal data (personal sensing, self reports, and LLM conversational logs) with a focus on data transparency, informed consent, and user trust. - Led qualitative research and prototyping efforts to develop provider dashboard interfaces that integrate multi-modal data, enhancing clinical reasoning and revealing longitudinal behavioral insights.	Cornell University (Remote)
2025	Research on Multi-modal Narrative Dashboard for Mental Health Advisor: Dr. Xuhai Xu Research Focus: Exploring how multimodal patient data can be shaped into clear, clinically meaningful narratives that improve clinicians’ insight discovery and decision making. - Co-designed MIND with five clinicians, creating a narrative-centered, text-based dashboard to visualize and synthesize multi-modal data (Clinical Data, Survey Scores, Passive Sensing). - Developed the full-stack MIND dashboard, combining rule-based analysis and LLMs to generate clinically grounded narrative insights from the multimodal patient data. - Directed the mixed-methods evaluation via a within-subject study (N=16 clinicians), leading the semi-structured interviews and qualitative analysis, alongside quantitative comparison surveys. Impact: Demonstrated significant improvement in insight discovery, decision confidence, and presentation efficiency. Resulted in a co-first-authored publication currently in revisions at CHI 2026.	Columbia University
2021 – 2022	Research on Peer programming in Computer Science Education Advisor: Dr. Steve Oney Research Focus: Exploring whether and how grouping novice programmers by the similarity of their code solutions affects the quality and outcomes of their subsequent discussions in peer learning. - Conducted a mixed-methods experiment (N=15) comparing similar vs. different code pairings via quantitative and qualitative analysis. Impact: Demonstrated that students paired with different solutions engaged in more descriptive conversations and produced more diverse follow-up coding solutions. Published as a first-authored paper at CHI LBW 2023.	University of Michigan - Ann Arbor

2020	Research on Smart Home Privacy Advisor: Dr. Yaxing Yao Research Focus: Investigating users' and bystanders' preferences for receiving privacy-related notifications in smart homes to address unclear data practices and raise awareness among multiple stakeholders. - Collaborated on a mixed-methods survey (N=259) comparing four awareness mechanisms (Dashboard, App, Lighting, Speaker) through effectiveness ratings and thematic analysis of contextual scenarios. Impact: Identified the Mobility App as most preferred and revealed conflicts between users' focus on usability vs. bystanders' concern for social norms. Published at CHI 2022.	Remote
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Professional Experience

2023 – Present	Moderna, Inc – <i>Software Engineer</i> • Contributed to front-end development of Moderna Direct's e-commerce platform using React, JavaScript, Redux, and Material UI, translating Figma designs into WCAG-compliant interfaces with backend API integration. • Collaborated cross-functionally with UX designers and engineers to refine requirements, resolve communication gaps, and ensure seamless integration of design and engineering workflows.	Seattle, WA
2022 – 2023	Zoom Video Communications, Inc – <i>Front-end Engineer</i> • Implemented essential features for Zoom's Chat platform, utilizing ReactJS, Typescript, SASS, and HTML5.	Remote

Professional Activities and Service

Reviewer

Journal	ACM Transactions on Computing for Healthcare (HEALTH)
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Teaching

2022	SI 579: Building Interactive Applications Graduate Student Instructor	University of Michigan Instructor: Prof. Ben Mullins
2021	SI 106: Programs, Information and People Graduate Student Instructor	University of Michigan Instructor: Prof. Steve Oney
2021	SI 579: Building Interactive Applications Graduate Student Instructor	University of Michigan Instructor: Prof. Steve Oney

Skills

Programming skills	Full-stack Development, React, Python, JS, TypeScript, CSS3, Git, Material-UI
UX Design skills	Figma, Adobe Creative Suite, Miro, and Sketch for prototyping
Research skills	Usability testing, Prototyping, Qualitative research methods and quantitative research methods.